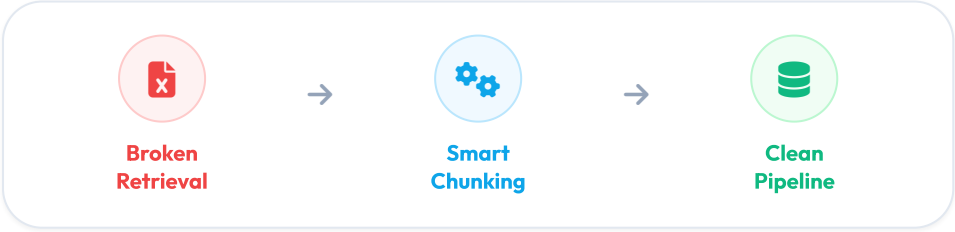
5

CHUNKING TECHNIQUES

That save your RAG pipeline in production

Stop using only **fixed chunking + overlap.**
Production RAG needs smarter strategies.



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TECHNIQUE 1

Fixed / Sliding Window Chunking

WHAT IT IS

Split text into fixed token/character sizes.

Usually:
500 tokens + 50 token overlap.

Most beginners start here.

HOW IT WORKS

INPUT

```
Chapter 1: Introduction to AI ...
Machine learning allows ...
Deep learning uses neural
networks ...
```

↓

OUTPUT

```
Chunk 1
Chapter 1: Introduction to AI ...
Machine learning allows ...

Chunk 2
Machine learning allows ...
Deep learning uses neural
networks ...
```

(Overlap keeps context.)

PRODUCTION WARNING

Can split **ideas mid-sentence**.
Semantic meaning may break.

FLOWCHART



WHEN TO USE

- Clean documentation
- Short blogs
- Consistent formatting

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TECHNIQUE 2

Semantic Chunking

WHAT IT IS

Chunk based on **meaning changes**, not token counts.

Sentence embeddings detect topic boundaries.

HOW IT WORKS

INPUT

```
Python is a programming language.
It supports object oriented
programming.

Embedding similarity drops here ↑

Stock markets react to inflation
news.
Interest rates affect investments.
```

↓

OUTPUT

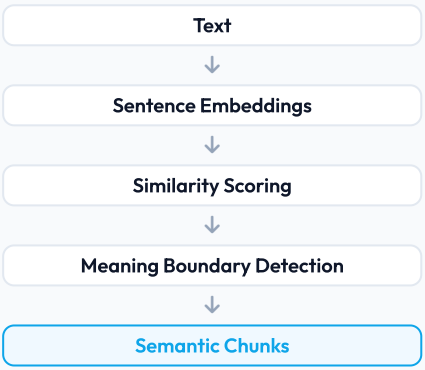
```
Chunk 1
Python programming section

Chunk 2
Finance section
```

PRODUCTION ADVANTAGE

Higher retrieval precision.
Less context pollution.

FLOWCHART



WHEN TO USE

- Knowledge bases
- Research papers
- Technical docs
- Multi-topic documents


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TECHNIQUE 3

Hierarchical Chunking

WHAT IT IS

Preserve **document structure**.

Chunk by:
Document → Section → Subsection → Paragraph.

Think tree hierarchy.

HOW IT WORKS

INPUT

```
API Guide
Authentication
  └ JWT Tokens
  └ OAuth
Rate Limits
  └ Requests per minute
```

OUTPUT

```
Level 1 Chunk
Authentication

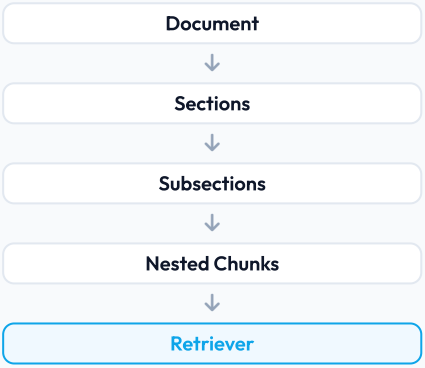
Level 2 Chunk
JWT Tokens

Level 3 Chunk
OAuth
```

PRODUCTION BENEFIT

Maintains parent-child relationships.
Better navigation.

FLOWCHART



WHEN TO USE

- API documentation
- Enterprise manuals
- Legal documents
- Large reports

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TECHNIQUE 4

LLM Based Chunking

WHAT IT IS

Let an LLM decide:
"Where should this document be split?"

Useful when data quality is messy.

HOW IT WORKS

INPUT

```
Messy OCR PDF
Invoice#123Paymentterms ...
Section missing ...
Broken tables ...
```

PROMPT

```
Split this text into logical
knowledge units.
```

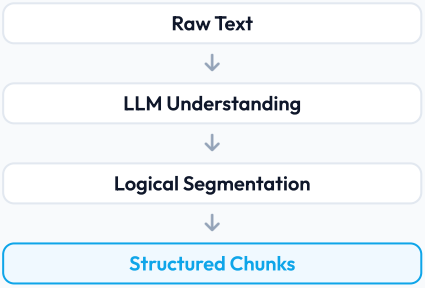
OUTPUT

```
Chunk 1
Payment Information
Chunk 2
Terms & Conditions
Chunk 3
Pricing Details
```

PRODUCTION WARNING

Higher latency.
Higher cost.
Use selectively.

FLOWCHART



WHEN TO USE

- OCR data
- PDFs
- Bad formatting
- Mixed layouts

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TECHNIQUE 5

Agentic Chunking

WHAT IT IS

Adaptive chunking.

Agent decides strategy dynamically.
No single splitter.

HOW IT WORKS

INPUT

```
Mixed Doc [Finance Tables + Clinical Text]
```

AGENT REASONING

```
LLM Analysis:
→ Finance data detected (needs precision)
→ Health text detected (needs semantic context)
```

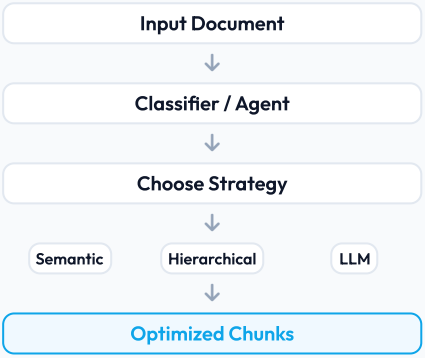
TOOL EXECUTION

```
Call Tool: Table_Chunker(Finance Segment)
Call Tool: Semantic_Splitter(Health Segment)
```

PRODUCTION BENEFIT

One pipeline handles many data types.
More scalable.

FLOWCHART



WHEN TO USE

- Enterprise RAG
- Multi-source ingestion
- Complex production pipelines

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COMPARISON SLIDE

Technique Comparison

TECHNIQUE	BEST FOR	COST	INTELLIGENCE
Fixed	Clean text	$ Low	Low
Semantic	Topic changes	$$ Medium	High
Hierarchical	Structured docs	$$ Medium	High
LLM	Messy data	$$$ High	Very High
Agentic	Enterprise systems	**Highest**	Adaptive

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FINAL TAKEAWAY

Final Takeaway

Most RAG failures are retrieval failures.

And retrieval failures often start with...

bad chunking.

CHOOSE CHUNKING BASED ON:

- Data structure
- Document quality
- Cost constraints
- Retrieval goals

Production RAG starts here.